Indian Institute of Technology, Kanpur
Proposal for a New Course

1. Course No: SPA***A

2. Course Title: Numerical Methods in Space Sciences & Engineering

3. Lectures per week: L=2, Tutorial: T=0, Laboratory: P=1, Additional hours: (0-2) =0 (A), Credits (3*L+2*T+1*P+A)=5
   Duration of Course: Half Semester

4. Proposing Department: Space, Planetary and Astronomical Sciences and Engineering.

5. Proposing Instructor: Pankaj Jain

6. Other Instructors who may teach this course: Sharvari Nadkarni-Ghosh, Kartick Sarkar, Rohit Sharma, Ishan Sharma

7. Course Description (A) Objectives: The course aims to introduce students to numerical techniques used in Space, Planetary and Astronomical Sciences and Engineering. (B) Contents (preferably in the form of 5 to 10 broad titles):

   1. **Numerical Integration** (2 lectures, 2 labs)
      Newton-Cotes formulae including trapezoidal rule and Simpson's rule, Gaussian quadrature, convergence and scaling of error, Monte Carlo Integration
   2. **Numerical Root finding** (1 lecture, 1 lab)
      Bisection method, Newton Raphson method for single and multi-dimensional systems.
   3. **Integration of Ordinary Differential Equations (ODE)** (3 lectures, 3 labs)
      Initial Value Problem Forward and backward Euler method, Runge-Kutta Method, Stiff system of equations, implicit vs. explicit schemes, shooting method for the Boundary Value Problem.
   4. **Numerical Interpolation of functions** (1 lecture, 1 lab) Numerical fitting with polynomials, Spline Interpolation

   (C) Pre-requisites, if any: N/A

   (D) Short summary for including in the Courses of Study Booklet: Numerical integration of functions, numerical root finding, Solving Ordinary differential Equations, Numerical Interpolation

7. Recommended Books:

   – Numerical Recipes: The Art of Scientific Computing
     Authors: Vertterling, Flannery, Press and Teukolsky.

8. Any other remarks:

Dated: Proposer:

Dated: DUGC/DPGC Convener:

The course is approved/not approved

Chairman, SUGC/SPGC

Dated: